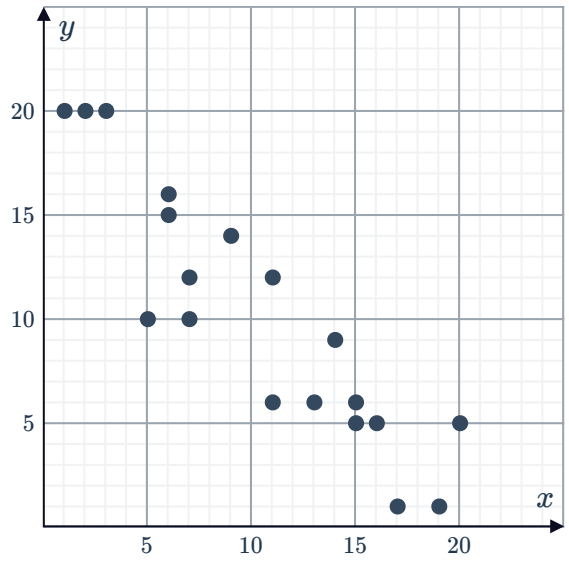
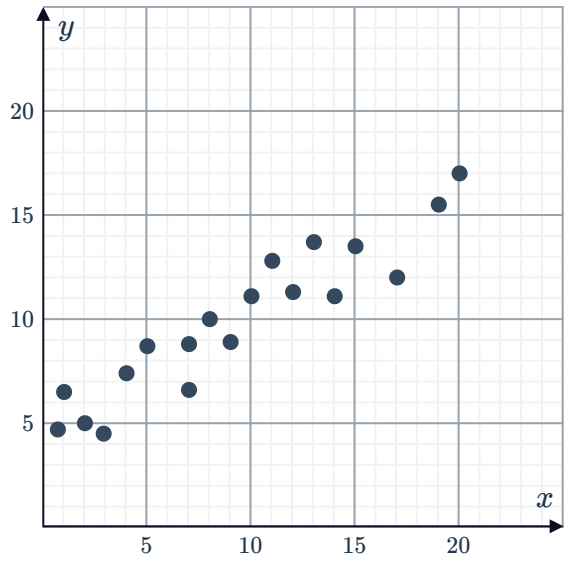
Line of best fit

1 Draw an approximate line of best fit by hand for each of the the scatter plots below:

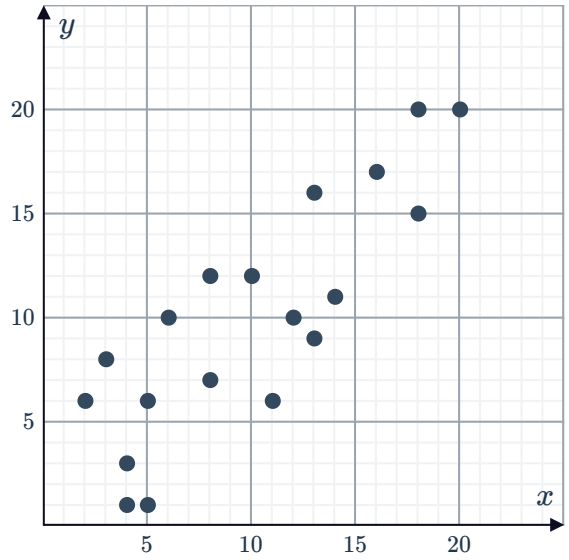
a



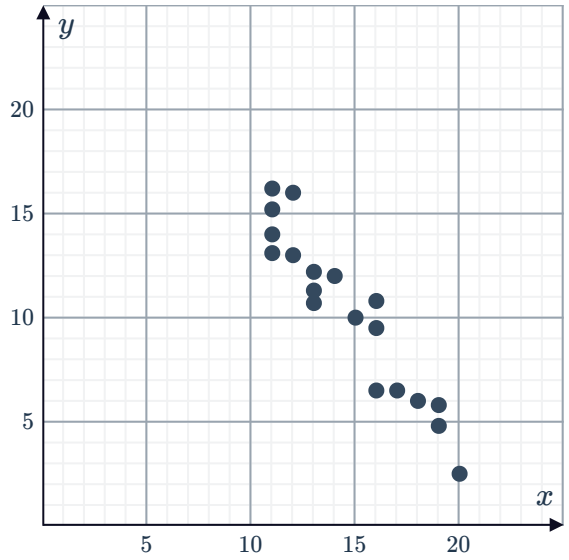
b



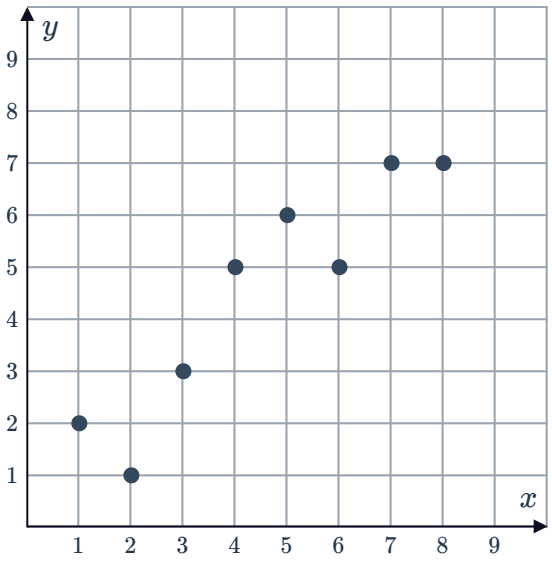
c



d



2 The following scatter plot shows the data for two variables, x and y:



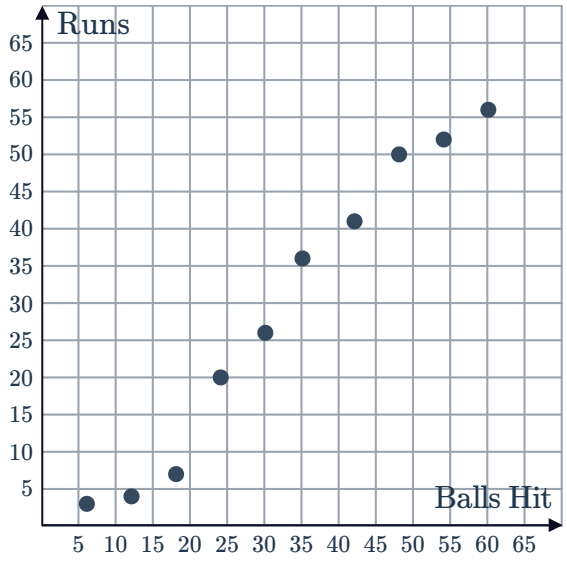
a Sketch the line of best fit for this data.

b Use your line of best fit to estimate the value of y when:

i $x = 4.5$ **ii** $x = 9$

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3 The following scatter plot graphs data for the number of balls hit and the number of runs scored by a batsman:



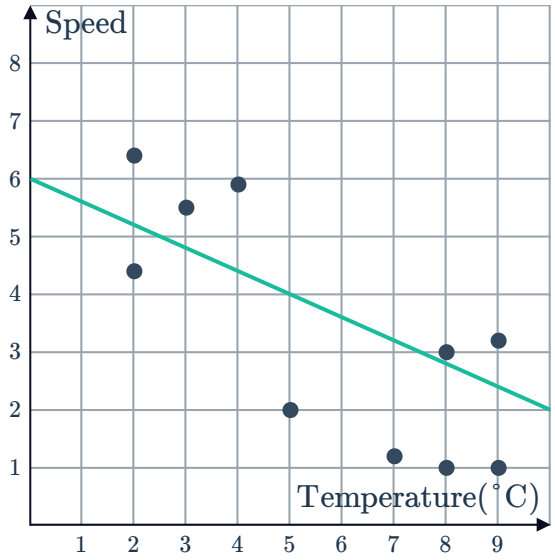
a Sketch the line of best fit for this data.

b Use the line of best fit to estimate the number of runs scored by the batsman after hitting:

i 27 balls **ii** 66 balls

c Is the relationship between the two variables positive or negative?

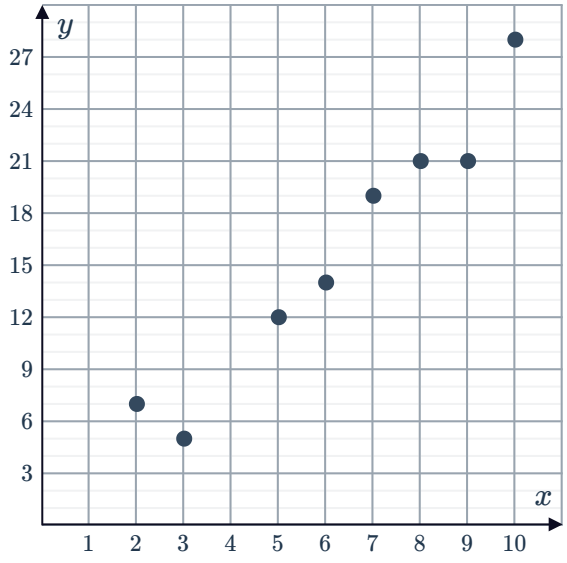
4 The average monthly temperature and the average wind speed, in knots, in a particular location was plotted over several months. The graph shows the points for each month's data and their line of best fit:



Use the line of best fit to approximate the wind speed on a day when the temperature is 5°C.

Equation of a line of best fit

5 Consider the following scatter plot:



a Is the relationship between the x and y variables positive or negative?

b Sketch the line of best fit for this data.

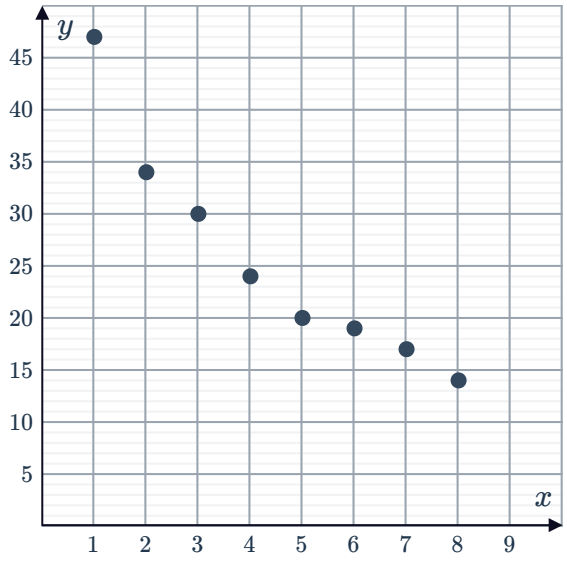
c Which of the following could be the equation for the line of best fit:

A $y = 2 - 3x$ **B** $y = 3x + 2$

C $y = -3x - 2$ **D** $y = 3x - 2$

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6 Consider the following scatter plot:



a Is the relationship between the x and y variables positive or negative?

b Which of the following could be the equation for the line of best fit:

A $y = -4x - 4$ **B** $y = 44 + 4x$

C $y = -4x + 44$ **D** $y = 4x - 4$

7 Use technology to find the line of best fit for the sets of data below. Write the equation with the coefficient and constant term to the nearest two decimal places.

a

x	24	37	19	31	32	22	14	30	23	40
y	-7	-8	-3	-6	-9	-8	-2	-8	-8	-12

b

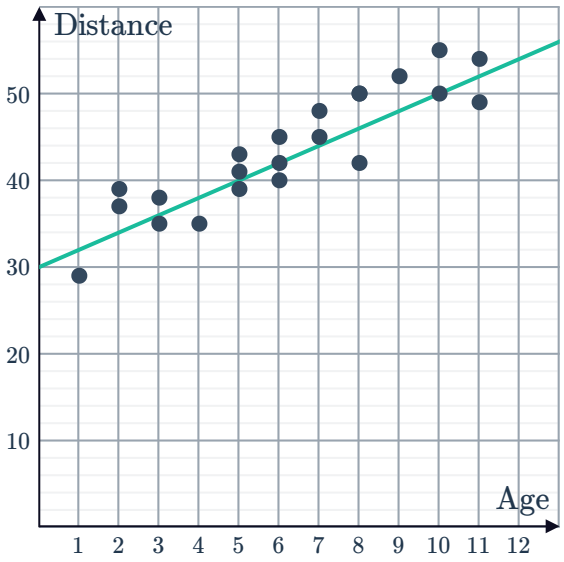
x	7	18	16	9	16	19	8	8	12	7
y	16.29	3.56	7.25	12.11	8.25	3.26	12.5	12.5	10.33	14.29

c

x	44	39	41	50	45	55	48	54	44	43
y	2.29	1.57	3.86	3.14	4.43	4.86	3.86	4.71	4.29	4.14

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8 Several cars underwent a brake test and their age, x (in years), was measured against their stopping distance, y (in metres). The scatter plot shows the results and a line of best fit that approximates the positive correlation:



a According to the line, what is the stopping distance of a car that is 6 years old?

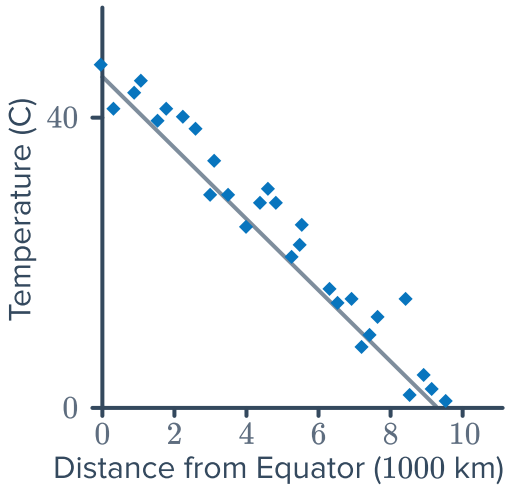
b According to the line, what is the stopping distance of a car that is 10 years old?

c Using the information found above, determine the gradient of the line of best fit.

d State the value of the vertical intercept of the line.

e Use the line of best fit found to estimate the stopping distance of a car that is 4.5 years old.

9 The distance of several locations from the equator and their temperature on a particular day is measured. The values are presented on the following scatter plot:



a Determine whether the following could be the equation of the line relating distance (x) and temperature (y):

A $y = -0.005x + 49$

B $y = -0.005x - 49$

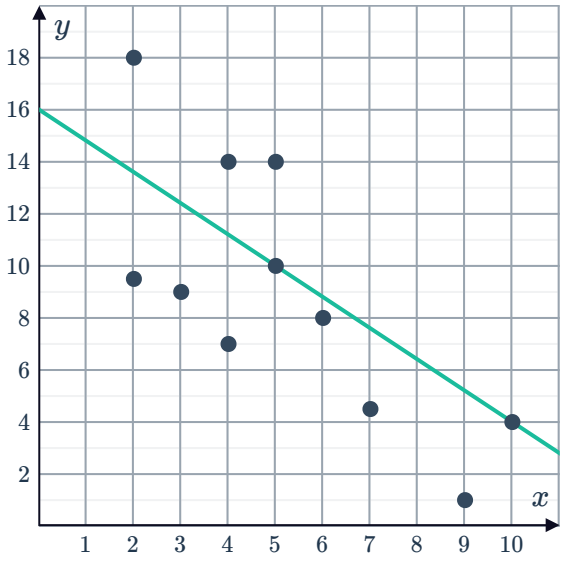
C $y = 0.005x + 49$

D $y = 0.005x - 49$

b Estimate the distance from the equator, x, if the temperature is 30.59°C.

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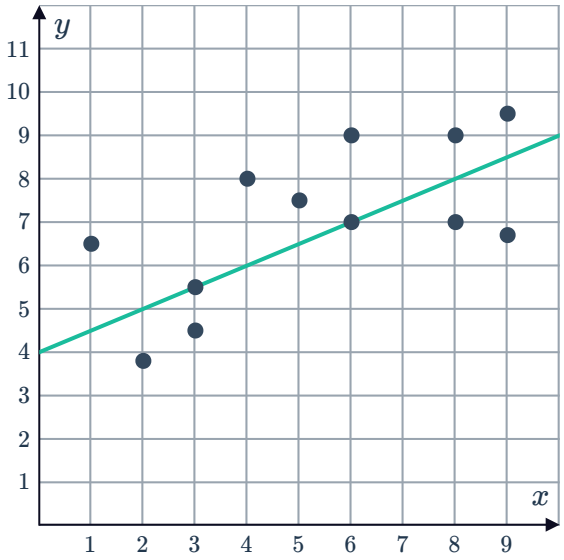
10 Find the equation of the line of best fit on the scatter plot shown:



11 Consider the scatter plot shown:

a Find the equation of the line of best fit.

b Use the line of best fit to approximate the value of y for $x = 6.9$.



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